

GDD

Treasure Archipelago



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Índice

Game Description	3
Game Mechanics	4
Progression	4
Mechanics	4
-Turns	4
-Base Stats	4
-Resources	4
-Points	5
-Dirtiness	5
-Ships life points	5
-Ration modes	5
-The map	5
-Sail modes	5
-The settlements	6
-Unit Roles	6
-The battle system	6
-Disease System	8
-Event System	8
Diseases	9
<i>Scurvy</i>	9
<i>Typhus</i>	9
<i>Typhoid</i>	9
<i>Malaria</i>	9
<i>Dysentery</i>	10
State of the Art	11
Educational Objectives	12
Visual Style	13
Interface	13
Technical Specifications	16

Game Description

Treasure Archipelago is a survival game set in an alternative version of the XVII century, during the golden age of piracy, where rumors about an archipelago full of treasure have been spreading around and every man around wants a piece of it. The player is no exception, but with a small old boat and not many men boarding it, the journey will prove difficult, and most likely deadly. But all the greatest men of history had to take risks for glory!



Game Mechanics

Progression

The player starts with a small vessel and a small crew of 10 pirates, with the goal of finding the treasure archipelago. Along the way, the player will encounter different vessels, some, foes like pirate ships, merchant ships and navy ships that the player can fight in order to plunder their riches.

The player might also stop at islands that enable the player to sell treasures, buy supplies and consumables, recruit inhabitants which may be interested in joining the crew, upgrade the ship, and allow their crew to have fun and rest on local inns.

While traveling the seas, the player will have to manage his crew, paying attention to each member's physical and mental well-being, as well as manage his resources, such as food and water. The crew will become sick throughout the journey by different reasons, such as lack of clean water or vitamin D. If a sick crew member is left unattended for too long, he may die.

The objective of the game is to reach the treasure archipelago while taking all the riches for yourself, preferably with a large ship and a crew strong enough to be able to stop anyone trying to do the same.

Mechanics

-Turns

The game is turn based so everything is calculated on a turn per turn basis, the crew consumes resources at the end of each turn. Actions taken by the player like moving forward or cleaning/repairing the ship take one turn.

-Base Stats

Each crew member has his own stats (physical health, mental health, hunger, vitamin C and thirst) if the health stats reach 0 the crew member dies.

There are also global stats which are influenced by the crew members individual stats, and the player choices (sail points, fight points, cleanliness).

If all the crew members die the player loses the game.

-Resources

Consumable resources (Food, fresh food, water and rum) are necessary to keep the crew healthy and happy, they will keep the crew members stats from deteriorating, will be consumed along the journey and need to be replenished. Not having these resources for too long will have very negative effects including but not limited to loss of mental and physical health, diseases and death from starvation or dehydration.

-Points

Fight points(fp) and sail points (sp) are used in battles against other ships. Both of them are calculated based on the number of crew members and their upgrades (that can be purchased on blacksmiths) and both serve an important role during battle .

-Dirtiness

Dirtiness is an important factor in preventing diseases, the more dirty the ship is the more likely it will be for diseases to spread, the ship starts off fairly clean but it will get dirtier as the game progresses, the player can take a turn to do maintenance on the ship which includes cleaning it.

-Ships life points

Life points are an important factor in winning battles, if they get to 0 the player loses the game. The ship starts out with the maximum amount of life points but battles can and most likely will drop them down. The player can take a turn to do maintenance on the ship which includes repairing it. The maximum amount of life points can be increased in the blacksmith.

-Ration modes

The player can choose between 3 ration modes (normal, ration and feasts) for food and water separately. In the normal mode each pirate will consume the normal amount (1 food/water per turn) and won't lose health, in the feast mode each pirate will consume twice the normal amount and will gain health, in the ration mode each pirate will consume half the normal amount but will lose health.

-The map

The map displays the properties of the place the player is currently on, it can be open sea, desert island, open sea with a battle, desert island with a battle or settled islands, the place where the player influences what he is able to do, the kinds of disease the pirates catch and the events that may happen, in the map the player can press the forward arrow to pass the turn and move forward.

-Sail modes

The player can change between two sail modes, trade routes and high seas, the sail mode affects the places the player goes to. In the trade routes sail mode the player will be able to find settled islands and will find easier battles that don't provide much rewards, making it the low-risk/low-reward sailmode. In the high seas sail mode the player won't find any settled island (making him unable to repair the ship or buy resources while in this sail mode) and will find more and harder battles with good rewards making it the high-risk/high-reward sailmode.

-The settlements

Settlements are essentially trade centers, each settlement has a marketplace, a blacksmith and a tavern. The player can buy different types of resources, consumable items that can be used in battle and sell treasures in the marketplace. Upgrade the sailing tools, weapons, cannons and hull of the ship in the blacksmith. Use the brothel to cure mental health of the crew, use the inn to cure the physical health of the crew and recruit new crew members in the tavern.

There is always the risk of their crew getting infected by some local disease.

Depending on the ship, the player will need a certain number of sailors to be able to travel. If for some reason the ship doesn't have enough sailors to keep it sailing, it may sink and cause a loss for the player. Having more sailors than necessary will provide bonuses to the sailing speed.

Random events may also occur depending on the crew's and ship's overall condition.

Each crew member also has a profession which enables certain actions or has certain effects, listed below.

-Unit Roles

- **Sailors**

keep the ship moving and working, preventing it from sinking by providing sail points, the amounts of sail points they give can be improved by upgrading the sailing tools. In battle sailors will try to evade enemy cannon fire and position the ships in a favourable position they can also be ordered to fight and provide fight points, but will lose mental health when doing so;

- **Fighters**

are trainers warriors that excel at boarding other ships and defending the ships from enemies by providing fight points, the amounts of fight points they give can be improved by upgrading the weapons. Fighters also help sailing the ship by providing a small amount of sail points;

-The battle system

The battle system is based on two stages: the fire stage and the sword stage. These two stages succeed each other until the battle is over, which happens when the player manages to successfully flee or either ships life points goes to 0 or below .

- **The fire stage**

in this stage the two ships are far from each other and will trade cannon fire while trying to evade hostile shots. The player has two choices in this stage to trade fire with the enemy or to escape the fight.

When trading fire both the player and the enemy will fire at each other, which can lead to 3 results: the opposing ship avoiding all the shots resulting in 0 damage being dealt, landing a regular hit resulting on the opposing ship resulting in damage equal to the cannon fire stat of the firing ship being dealt to the opposing ship, landing a direct hit resulting on the opposing ship resulting in damage equal to twice the cannon fire stat of the firing ship being dealt to the opposing ship. The changes of the outcome is based on both ships sail points (the more sp you have the most likely you'll be to land your shots and avoid the enemy shots). The player can upgrade their cannon fire stat by buying cannon upgrades in the blacksmith.

When escaping the player can either succeed or not (change based on players sp and enemy sp), if the player is successful the battle will simply end without any reward being gained, if not the player will take a direct hit as punishment and the battle continues as normal.

- **The sword stage**

in this stage the two ships are close from each other, the crews will try to board the opposing vessel. The player has two choices in this stage to fight the enemy or to disengage.

When fighting both crews will fight each other, one side will win and board the opposing ship and deal heavy damage to it (chances of winning and damage dealt are based on fight points). Every crew member that fought will lose physical health, they take more damage if the player loses the fight, if sailors fight they will use mental health as well as physical health.

When escaping the player can either succeed or not (change based on players sp and enemy sp), if the player is successful he will take no damage and move to the fire phase, if not the player will take heavy damage as punishment and battle continues as normal.

Battles always start in the fire phase. There are two types of consumable items that can be the marketplaces and used in battle: explosives cannon shells that can be used in the fire phase and will double your cannon damage and adrenaline drugs can be used on the sword stage to increase your fp and damage dealt to the enemy ship in case of victory but it will take on toll on the user mental health. There are 3 types of enemies (merchants, pirates and navy ships), each enemy gives a different rewards if defeated, the type of the enemy found will depend on the sail mode the player has chosen and the strength of said enemy will depend on how many turns have passed since the beginning of the run.

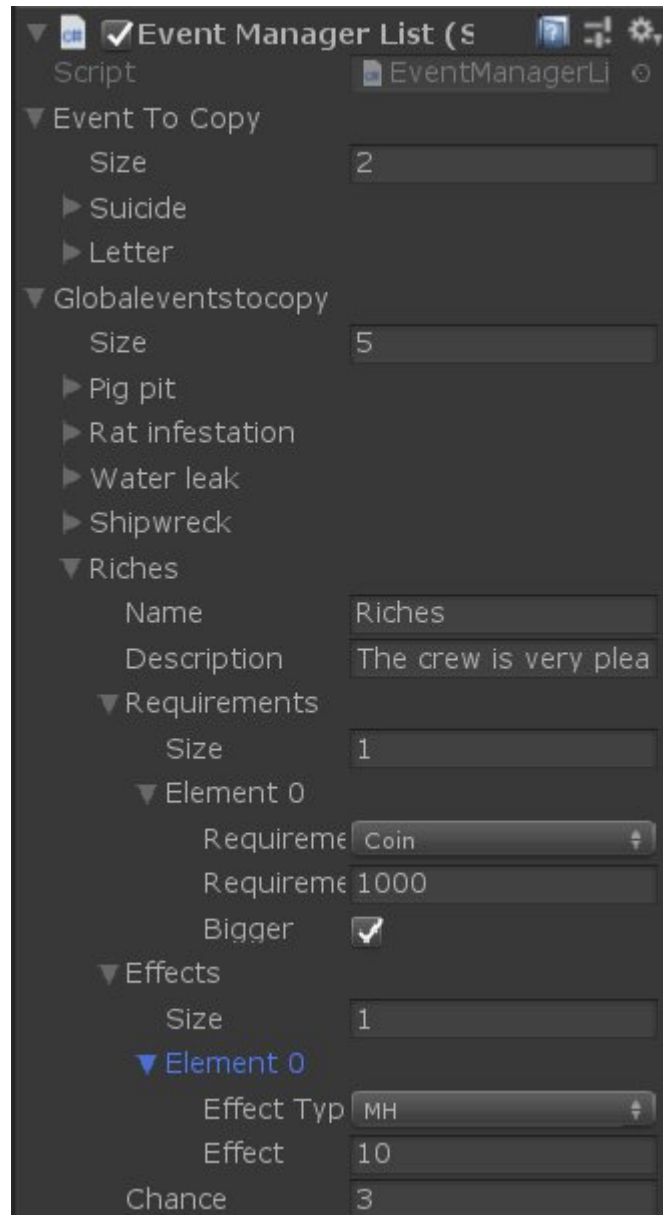
-Disease System

The disease system works every turn by first rolling any new diseases respecting any requirements if there are (for example Malaria can occur whenever the player visits an island, with extra chance added depending on the dirtiness state of the crew), and then running all disease logic depending on the disease (each disease has its own unique triggers and effects, including effects that make the pirate consume more water, or increase the dirtiness factor).

-Event System

The event system works somewhat like the disease system by first rolling any events (and to even be chanced some events need to have all their requirements met), and then applying their effects. There are negative and positive events, affecting individual pirates, like a pirate getting a letter from home which increases his sanity, or coming suicide (which can occur only on very low sanity), and events which have global effects like a water leakage reducing the water supply, or a global sanity boost on all pirates when there is a lot of gold present.

There is also an easier interface built on the unity inspector which allows the creation/alteration/deletion of any events without writing any code, since ideally there would be a huge amount of events, and writing each effect and requirement by code would be a very time consuming task.



Diseases

Scurvy

A disease resulting from a lack of vitamin C (ascorbic acid). Early symptoms include weakness, feeling tired, and sore arms and legs. Without treatment, decreased red blood cells, gum disease, changes to hair, and bleeding from the skin may occur.

Cause - Lack of vitamin C.

Effect - Less effectiveness while working, lasting about 14d, may be treated with ingestion of vitamin C and is fatal.

Typhus

Common symptoms include fever, headache, and a rash. The diseases are caused by specific types of bacterial infection. Epidemic typhus is spread by body lice, scrub typhus is spread by chiggers, and murine typhus is spread by fleas.

Cause - unclean environment.

Effect - Less effectiveness while working and a decrease of mental sanity, lasting about 30d, may be treated by a medic, can cause death.

Typhoid

Often there is a gradual onset of a high fever, weakness, abdominal pain, constipation, headaches, and mild vomiting also commonly occur. Other people may carry the bacterium without being affected, however, they are still able to spread the disease to others.

Cause - unclean environment

Effect - Less effectiveness while working and a decrease of mental sanity, lasting about 30d, may be treated by a medic, sometimes causes death.

Malaria

Malaria causes symptoms that typically include fever, tiredness, vomiting, and headaches.

Cause - Mosquito bites

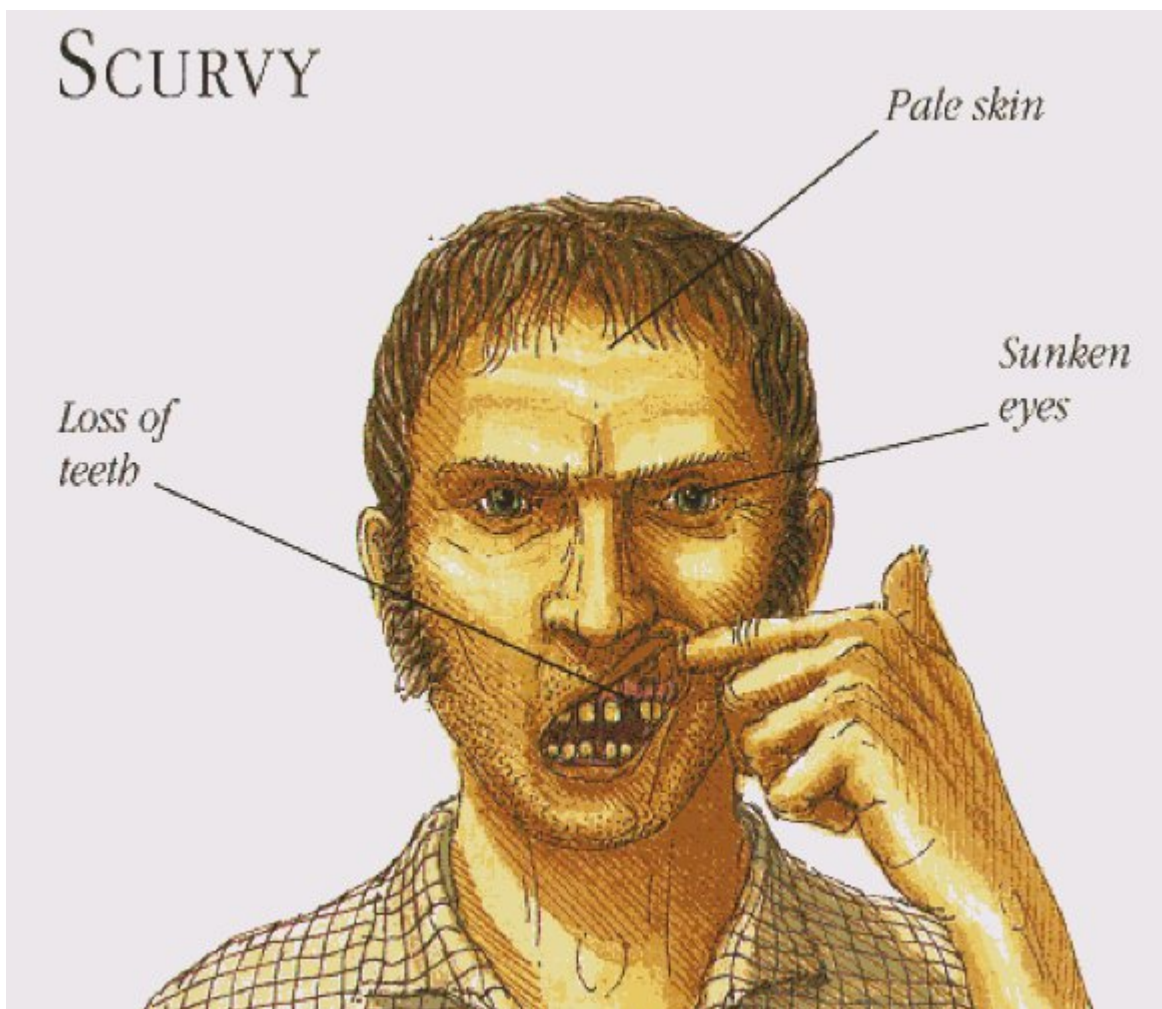
Effect - Less effectiveness while working and uncleanliness. Lasting about 30d, may be treated by a medic, can cause death.

Dysentery

An inflammatory disease of the intestine, especially of the colon, which always results in severe diarrhea and abdominal pains. Other symptoms may include fever and a feeling of incomplete defecation.

Cause - Uncleanliness

Effect - Uncleanliness and increased water consumption. Lasting about 7d, may be treated by a medic and rarely causes death.

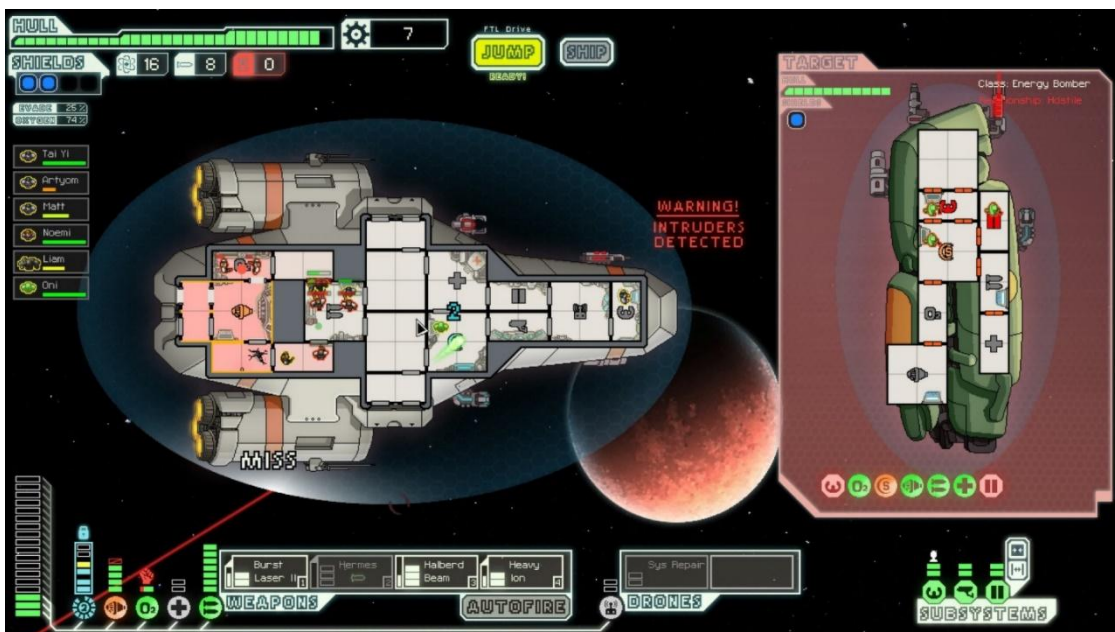
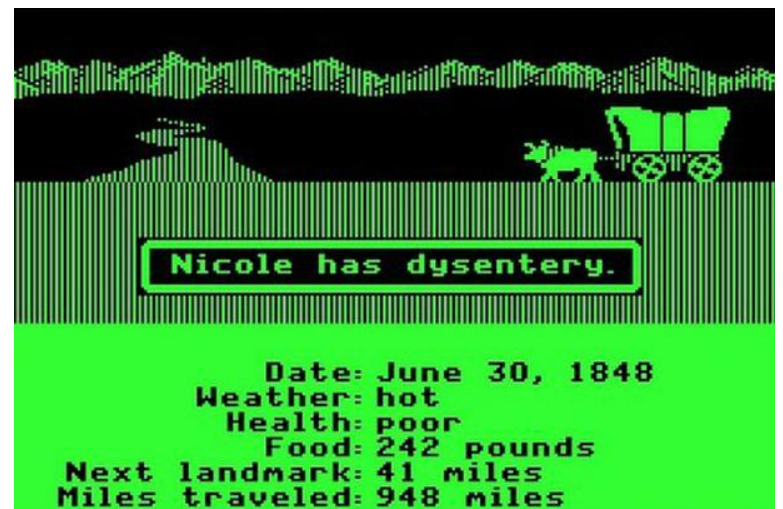


State of the Art

The management of the physical and mental well-being of various distinct crew members aspect of the game was inspired by “Oregon Trail”, an educational computer game from the early 70s. The exploration, ship encounters and random events were inspired by the indie game “Faster than light”.

We plan to offer an original outlook on the conditions and diseases faced by the sailors at that time, which is, as far as we know, an unexplored concept in video games and we think it has enough potential to make our game unique, interesting and educational.

Oregon Trail a game where the player leads a wagon across the Unites States, managing resources and the health of a group of travelers



FTL: FASTER THAN LIGHT a game where the player manages a spaceship and crew travelling through space, facing enemies along the way

Educational Objectives

With this game we have two objectives: To provide our players with an interesting experience, and to educate them on the diseases, health conditions and treatments faced by the people of the time, which in some cases are still relevant today.

We plan to do that by accurately representing the diseases with different in-game mechanics and by providing descriptions of the symptoms and cures on the in-game events. We also hope to entice our players to do further research on the diseases by “showing them in action” in our game.

The player can also a in-game catalog that describes all the diseases their crew has caught during the current run and a in-game quiz about said diseases where the player can test his knowledge about the subject



Visual Style

This game will exclusively use 2D pixel art style, original, drawn by us.

We chose 2D pixel style over other styles like 3D modeling since it would require us to make fully functional models, which take a lot of time to create, and unity more frequent than not ends up not being fully compatible with the model. Art-wise, 3D modeling doesn't achieve anymore for our game than what 2D sprites can achieve, so the choice became clear to us.

Interface

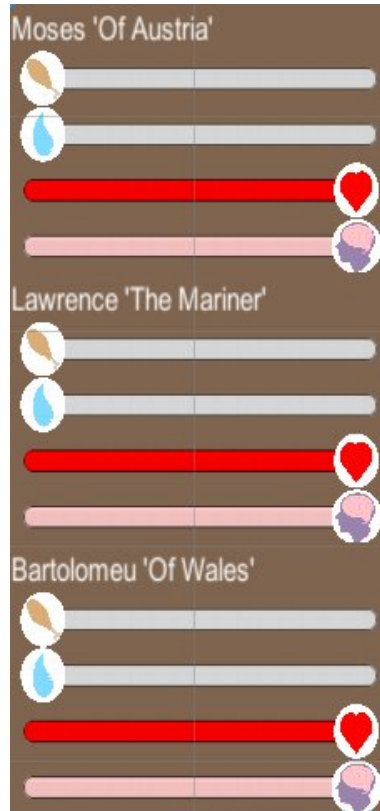
Interface for the main screen



The resource bar allows the player to see their resources and money with a quick glance.



The status bars give a simplified overview of the crew member's health.



This button will allow the player to access his inventory where treasures that can be sold in stores and consumables that can be used in battle are stored



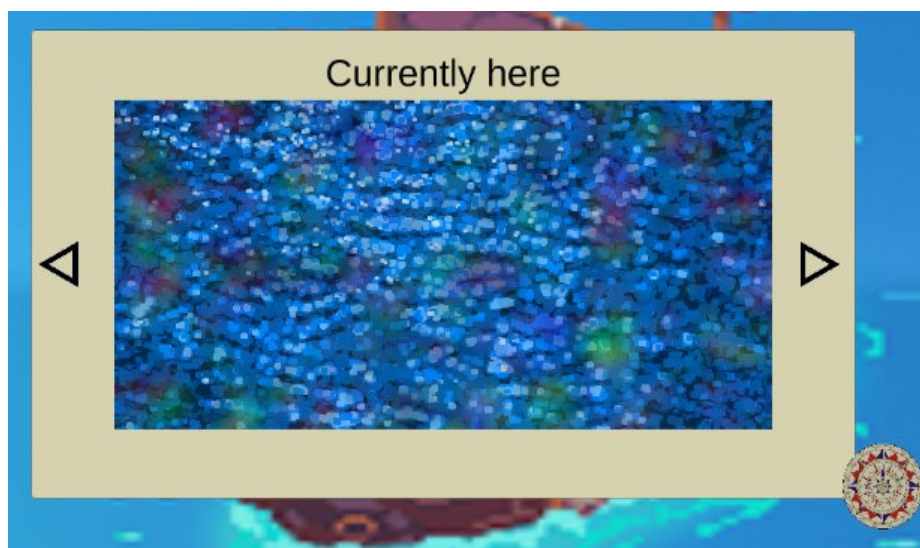
This buttons lead to the in-game disease catalog and quiz.



This button will lead the player to a screen where it's possible to change the sail mode, the ration modes, and see the life points of the ship, the ship maintenance button and the log registering the quantity of resources consumed on the previous turn



The compass button opens the map where the player can move forward or check the places that have been previously visited.



Technical Specifications

Game engine

We will be using the Unity3D game engine since it works well for small scale games, it also has the Unity Collab feature which allows projects to be easily shared between small teams, and we have prior experience with it.

Platform and OS

Unity gives the possibility to export the game to various platforms, including: iOS, Mac Standalone, Windows Standalone, Web, Nintendo Wii, Xbox 360, PS3, and Android. This game specifically will be developed for Windows, but it could be easily ported to android and iPhone with a few adjustments.

Coding

We will be using the c# language for scripting.

